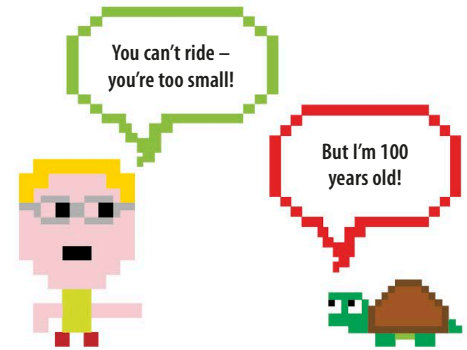


Ride the rollercoaster

A sign at the theme park says you must be over 8 years old and taller than 1.4 metres to ride the rollercoaster. Mia is 10 years old and 1.5 metres tall. Let's use the shell to check whether she can go for a ride. Type the following lines of code to create variables for Mia's age and height and assign the correct values to them. Type the rules for going on the rollercoaster as a Boolean expression, then hit the enter/return key.



These two lines assign values to the variables.

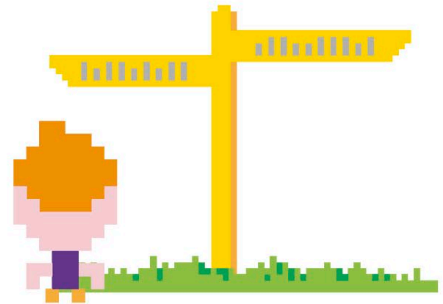
```
>>> age = 10
>>> height = 1.5
>>> (age > 8) and (height > 1.4)
True
```

Mia can go on the rollercoaster!

This is a Boolean expression meaning "older than 8 and more than 1.4 metres tall".

Branching

Computers often need to make decisions about which parts of a program to run. This is because most programs are designed to do different things in different situations. The route through the program splits like a path branching off into side paths, each leading to a different place.



LINGO

Condition

A condition is a Boolean expression (a True-or-False comparison) that helps a computer decide which route to take when it reaches a branch in the code.

▷ School or park?

Imagine you have to decide what route to walk each day based on the answer to the question "Is today a weekday?" If it's a weekday, you take the route to school; if it's not, you take the route to the park. In Python, the different routes through a program lead to different blocks of code. A block can be one statement or several, all indented by four spaces. The computer uses a test called a condition to figure out which blocks it should run next.

